

TSUYAMA, Japan

WMO: 477560

Lat: 35.0648N

Lon: 134.0090E

Elev: 485

StdP: 14.44

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.0	26.8	19.0	14.6	34.4	21.1	16.1	35.0	18.6	37.3	16.2	37.9	3.5	270	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	92.4	77.4	90.2	76.8	88.0	76.0	79.2	88.4	78.2	86.8	77.3	85.2	6.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	142.5	82.6	75.9	137.9	82.1	75.0	133.7	81.5	43.1	88.9	42.1	86.9	41.1	85.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	11.9	9.9	21.0	95.5	2.9	1.9	18.9	96.9	17.2	98.0	15.6	99.1	13.5	100.5		
(5)			20.3	81.0	3.0	1.0	18.2	81.7	16.5	82.3	14.8	82.8	12.7	83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	37.2	39.0	45.0	54.9	64.2	71.3	78.7	80.7	73.0	61.9	50.5	41.0
	DBStd	15.83	3.82	4.79	5.51	6.07	4.45	4.24	4.06	3.41	5.22	5.49	5.77	4.64
	HDD50	1253	398	311	174	23	0	0	0	0	0	1	64	282
	HDD65	3900	863	729	620	306	68	3	0	0	4	127	435	745
	CDD50	4255	0	2	20	169	441	639	890	952	691	370	79	2
	CDD65	1426	0	0	0	2	44	192	425	487	245	31	0	0
	CDH74	11426	0	0	0	66	441	1149	3542	4538	1563	127	0	0
	CDH80	3921	0	0	0	7	68	260	1314	1837	426	9	0	0
Wind	WSAvg	3.8	4.2	4.2	4.4	4.2	4.0	3.5	3.6	3.7	3.5	3.3	3.3	3.9
Precipitation	PrecAvg	59.1	3.5	3.3	4.2	4.3	5.8	6.7	8.2	4.8	7.0	4.2	3.6	3.6
	PrecMax	79.4	5.8	4.7	7.6	7.1	12.2	11.8	14.9	13.2	17.2	9.4	8.0	5.8
	PrecMin	40.8	2.0	1.9	2.2	1.3	2.2	1.9	1.6	1.6	2.2	1.5	1.2	1.3
	PrecStd	8.4	1.2	0.9	1.3	1.8	2.6	2.3	3.6	3.0	3.9	2.2	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	61.1	69.7	80.5	85.4	89.0	95.0	95.8	90.5	80.8	69.7	58.3
		MCWB	45.2	50.1	54.2	60.6	64.7	73.3	78.7	78.0	75.8	70.1	59.9	51.1
	2%	DB	49.8	55.6	64.6	75.3	81.7	85.4	91.9	93.1	87.3	77.2	66.3	55.3
		MCWB	43.8	47.0	51.8	58.2	63.5	71.1	77.8	77.5	74.5	66.4	58.1	48.8
	5%	DB	47.4	51.7	60.4	71.5	79.0	82.9	89.6	91.2	84.8	74.7	63.4	52.3
		MCWB	41.5	44.6	49.7	56.8	62.7	69.7	77.1	77.0	73.2	64.8	56.5	46.4
	10%	DB	45.0	48.5	56.6	67.8	76.0	80.6	87.3	89.1	82.2	72.0	60.5	49.6
		MCWB	39.8	42.5	48.5	55.2	61.8	69.2	76.5	76.6	71.9	63.2	54.9	44.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	54.4	57.6	65.4	70.7	76.9	80.7	80.7	78.2	73.3	63.6	54.6
		MCDB	50.5	58.8	63.3	71.3	76.0	83.6	90.5	90.7	86.6	77.3	66.1	55.7
	2%	WB	44.8	48.9	54.8	62.3	67.9	75.0	79.2	79.4	76.6	70.0	60.5	50.0
		MCDB	47.9	52.9	59.7	68.7	73.7	80.9	88.5	88.8	83.7	74.4	63.6	53.4
	5%	WB	42.3	45.6	52.3	60.0	66.2	73.5	78.2	78.4	75.4	67.2	58.0	47.4
		MCDB	46.2	50.3	58.0	66.9	72.7	78.6	86.8	87.2	81.3	71.6	61.8	51.0
	10%	WB	40.3	42.7	49.2	57.9	64.4	71.8	77.2	77.6	74.1	64.7	55.6	45.2
		MCDB	44.4	47.8	55.9	64.5	72.2	77.1	85.1	85.8	79.4	69.6	59.5	49.1
Mean Daily Temperature Range	5% DB	MDBR	15.0	16.9	20.0	22.3	21.0	16.3	14.4	16.1	15.8	17.7	17.3	15.3
		MCDBR	18.6	23.0	27.6	29.1	27.7	21.0	18.2	19.4	18.2	20.2	19.6	18.9
		MCWBR	12.3	14.5	15.7	14.1	12.2	8.9	7.0	6.9	7.4	10.1	11.6	12.5
	5% WB	MCDBR	17.1	19.7	21.7	20.5	18.9	14.4	16.2	16.6	14.5	14.4	15.6	16.7
		MCWBR	12.2	13.8	14.8	11.7	10.7	7.2	6.8	6.5	6.5	8.6	10.6	12.2
Clear-Sky Solar Irradiance	taub	0.421	0.461	0.529	0.554	0.555	0.576	0.576	0.547	0.496	0.451	0.429	0.413	
	taud	2.037	1.957	1.821	1.819	1.873	1.893	1.952	2.024	2.123	2.151	2.113	2.081	
	Ebn at Noon	235	241	235	238	240	234	233	237	243	242	232	229	
	Edh at Noon	42	51	63	67	64	62	59	54	46	42	39	38	
All-Sky Solar Radiation	RadAvg	635	848	1182	1510	1660	1521	1554	1610	1248	1020	731	573	
	RadStd	47	79	98	131	157	127	219	173	118	96	61	44	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page